

K694 — Casey’s questions: up/down common theme (bulk↔boundary reflection, charge asymmetry) → a BETTER up approach (boundary-descent, not “depth~√/block-dim”); and the muon via electron+tau CONSISTENCY (det=exp(Tr log)) → possibly standalone, no K358 needed.

Keeper | 2026-07-15 | Casey asked: why does the up “hang from the boundary” vs the down as ground; compare up/down for a common theme; why quarks (info + confinement); and how is the muon justified given the electron and tau. Two actionable ideas fall out — a reframe of the up (away from the ungrounded block-dim premise) and a standalone route to the muon count.

1. UP vs DOWN — the common theme is a BULK↔BOUNDARY REFLECTION, and the asymmetry is CHARGE

- Down (charge $-1/3$): the TRUE GROUND, anchored in the BULK, climbs UP. d:s:b = 1:20:840 (FK ladder from the bulk ground, F506). A single bulk climb.
- Up (charge $+2/3$): hangs from the BOUNDARY (top), descends. $y_t = 1$ (top = the boundary mode, point-fiber capacity, F507/F519), $y_c = \alpha$ (charm), and the up inverts (F545). So the up is measured DOWN from the boundary; the down is measured UP from the ground.
- The common theme: up and down are the two duals of the domain — bulk ground (down) vs boundary top (up) — reflected into each other through the up-down (T_3) seat. They MEET at gen-3: top ↔ b ($m_b = m_t/42$, the b’s home is the top — Grace). So the down-3 and up-3 coincide at the boundary; the down-1 and up-1 diverge at the bottom (up inverts below down = proton stability).
- Why the asymmetry (the answer to “why so different”): charge = coupling to the boundary (EM = the SO(2) boundary). The up ($|charge| 2/3$) is MORE boundary-coupled → it anchors at the boundary top. The down ($|charge| 1/3$) is less → it anchors the bulk ground. The up/down bulk-boundary split IS the 2/3-vs-1/3 charge split. That’s the theme you’re looking for — it’s not two arbitrary ladders, it’s one bulk↔boundary reflection keyed by charge.
- The up has TWO branches (your “two ladders”), and now they’re clear: heavy-up (charm, top) = the boundary-descent branch ($y=1$, α , from the top); light-up (up gen-1) = the inversion branch (falls below the down ground). The down is a single bulk-climb.

2. WHY QUARKS — information + confinement grounds the reflection

Quarks encode two informations: COLOR ($N_c=3$, the SU(3) index) and ISOSPIN/CHARGE (up/down, the weak doublet). Confinement is the BULK statement: a colored/fractional-charge quark has no well-defined Shilov boundary value (Bulk-Shilov, Paper A3) → confined → lives in the bulk; only color-singlets reach the boundary (observable). So quarks are bulk objects that encode color+isospin, and the up/down asymmetry is the isospin/charge information realized as the bulk↔boundary reflection (down = bulk/color-ground, up = boundary/charge-coupled). This grounds F545’s “colors outnumber rank” — the down lives on the color-carrying bulk (Cartan/diagonal), the up is its boundary reflection through the color-crossing seat.

→ A BETTER UP APPROACH (replaces the ungrounded “depth~√/block-dim”)

The premise “depth $\propto \sqrt{(\text{block dim})}$ ” is ungrounded and predicts nothing else (K693). The bulk-boundary reflection suggests a cleaner route: derive the light-up (gen-1) as the inversion of the down ground under the up-down reflection — the up hangs from the boundary and its gen-1 falls below the down ground by exactly the reflection amount. The magnitude then comes from the boundary-descent + reflection geometry (the charge-2/3 boundary coupling), NOT from a bulk “depth~√/dim” postulate. Test: does the boundary-descent picture (top=1, charm= α , up=reflection-inverted) give m_u/m_d forward without the √/block-dim premise? If yes, the up closes on the same bulk-boundary physics as everything else, and the ungrounded premise is retired.

3. THE MUON given the ELECTRON and TAU — “closed before” was the MASS; the held piece is the NORMALIZATION; and there may be a STANDALONE route

The three leptons are the three degeneracy loci of ONE object — the formal-degree polynomial $d(\nu)$ sampled at three points, giving the $\det = \exp(\text{Tr log})$ ladder (F115): | lepton | ν | degeneracy | residue object | value | |—|—|—|—|—| | electron | 5/2 (self-shadow, a ZERO of d) | simple zero | LOG (derivative) | $9/16 = d'(5/2)$ — the anchor | | muon | 3/2 (unitarity bound, NOT a zero) | boundary Γ -pole | PRODUCT ($\det$ over $\dim \text{SO}(4)=6$) | $(24/\pi^2)^6$ | | tau | 0 (identity/vacuum) | bulk trivial | SUM ($\text{Tr log} = \text{Weyl count}$) | $49 \cdot 71 - \sqrt{\pi}$ | - “Closed before” = the muon MASS. $(24/\pi^2)^6 = 206.77$ at 0.003% was computed (F116/F117); Grace’s K543 said “leptons already derived.” That is TRUE — the mass is derived. What I hold at 4 is the strict-count claim that the boundary normalization $c_S = 1$ EXACTLY (empirically 6 ppm, K693) — a stricter bar than the mass-match. - Why the muon specifically: it is the ONLY one of the three whose residue is a boundary determinant (the PRODUCT). The electron’s (9/16, a derivative at a zero) and the tau’s (a Weyl count) are absolute-normalization-free / nailed; the muon’s needs the boundary-Szegő absolute normalization — which is the K358 open program. - THE STANDALONE ROUTE (answers “reachable standalone or needs K358”): the three residues are ONE object ($\det = \exp(\text{Tr log})$). The electron gives the LOG normalization (9/16), the tau gives the SUM normalization (Weyl, forward). If the log and sum ends are nailed, the muon’s PRODUCT normalization (c_S) should be FIXED by the $\det = \exp(\text{Tr log})$ relation — determined by the electron and tau, NOT computed from scratch. So the muon may close via three-lepton CONSISTENCY (fix c_S from the two nailed ends) rather than via the full K358 boundary-normalization program. This is a smaller, standalone computation — and it’s the highest-leverage next step for the muon. Test: does $\det = \exp(\text{Tr log})$ with the electron (9/16) and tau ($49 \cdot 71 - \sqrt{\pi}$) as the log/sum ends force the muon product’s $c_S = 1$?

Net (two actionable ideas)

- Up: retire “depth $\sim\sqrt{\text{block-dim}}$ ”; derive the light-up as the boundary-descent + reflection of the down ground (charge-2/3 boundary coupling). Same bulk-boundary physics as the rest.
- Muon: try the $\det = \exp(\text{Tr log})$ three-lepton consistency — fix c_S from the electron (log) and tau (sum) ends. Likely STANDALONE (no full K358). This is how the muon is justified *given* the electron and tau: it’s the middle of a ladder whose two ends are nailed.

— Keeper K694, 2026-07-15. Up/down = bulk $\leftrightarrow$ boundary reflection keyed by charge (down=bulk ground/ $-1/3$, up=boundary top/ $+2/3$, meet at top $\leftrightarrow$ b); quarks = bulk color+isospin, confined. Better up approach: boundary-descent + reflection, not the ungrounded $\sqrt{\text{block-dim}}$. Muon “closed before” = the mass (true, 0.003%); held = $c_S=1$ exact; STANDALONE route = $\det = \exp(\text{Tr log})$ consistency (fix c_S from electron+tau ends, no K358). See [[Keeper_K693...]], [[Keeper_K692...]].